

Day 21 - Lecture 21.11: The Cross Product

Python 3.10+

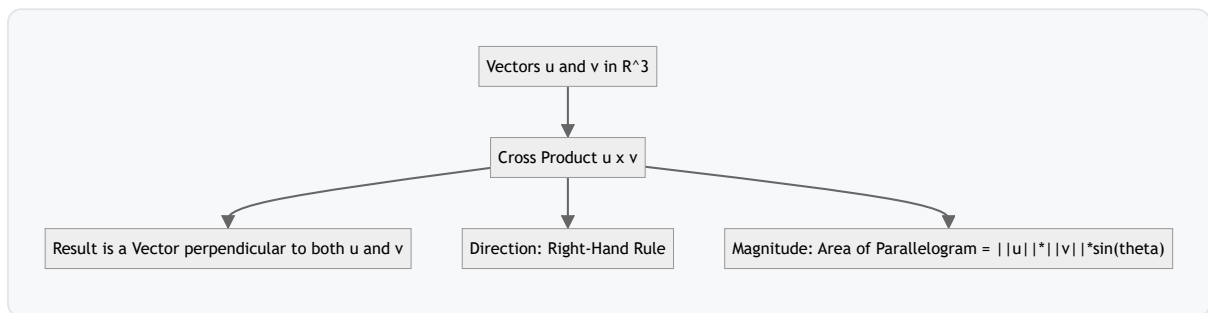
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1. Overview & 3D Spatial Geometry

While the dot product yields a scalar, the **Cross Product** of two 3D vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^3$ yields a new **vector** that is strictly perpendicular to both input vectors simultaneously. In Robotics, Computer Vision, and 3D Generative AI (NeRFs, Gaussian Splatting), cross products calculate surface normals, rotational torque, and plane orientations.



2. Mathematical Formalism

1. Determinant Expansion Definition:

For vectors $\mathbf{u} = [u_1, u_2, u_3]^T$ and $\mathbf{v} = [v_1, v_2, v_3]^T$:

$$\mathbf{u} \times \mathbf{v} = \det \begin{bmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} & u_1 & u_2 & u_3 & v_1 & v_2 & v_3 \end{bmatrix} = \begin{bmatrix} u_2v_3 - u_3v_2 & u_3v_1 - u_1v_3 & u_1v_2 - u_2v_1 \end{bmatrix}$$

2. Geometric Magnitude & Direction:

$$|\mathbf{u} \times \mathbf{v}| = |\mathbf{u}| |\mathbf{v}| \sin(\theta)$$

where $|\mathbf{u} \times \mathbf{v}|$ equals the exact **area of the parallelogram** spanned by $\mathbf{u}$ and $\mathbf{v}$.

3. Key Algebraic Properties:

- **Anti-commutative:** $\mathbf{u} \times \mathbf{v} = -(\mathbf{v} \times \mathbf{u})$
 - **Self Cross-Product:** $\mathbf{u} \times \mathbf{u} = \mathbf{0}$
 - **Parallelism Test:** $\mathbf{u} \parallel \mathbf{v} \iff \mathbf{u} \times \mathbf{v} = \mathbf{0}$
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3. Python Implementation

```
import numpy as np

u = np.array([1.0, 0.0, 0.0]) # Unit vector along X
v = np.array([0.0, 1.0, 0.0]) # Unit vector along Y

# Cross Product
w = np.cross(u, v)

print(f"u:      {u}")
print(f"v:      {v}")
print(f"u x v:    {w} (Expected: [0, 0, 1] along Z)")
print(f"Orthogonal to u: {np.dot(w, u) == 0}")
print(f"Orthogonal to v: {np.dot(w, v) == 0}")

# Parallelogram Area
area = np.linalg.norm(w)
print(f"Parallelogram Area: {area:.2f}")
```

4. Key Takeaways & Interview Points

- **Strictly in 3D:** Cross product is uniquely defined for vectors in $\mathbb{R}^3$ (and algebraically in $\mathbb{R}^7$).
- **Computer Vision Surface Normals:** Estimating surface mesh orientations requires computing cross products between adjacent triangle edge vectors.